

Registering a HIDS Agent with a SIEM Console

Executive Summary

Overview

This project demonstrates the successful deployment of multiple Host-Based Intrusion Detection System (HIDS) agents with a Wazuh-based Security Information and Event Management (SIEM) console. The Wazuh Manager was installed on an Ubuntu Server, the Dashboard GUI was hosted on an Ubuntu Desktop, and HIDS agents were deployed on both the Ubuntu Desktop (Agent 001) and a Windows Server 2022 system (Agent 003). Each agent was registered using secure authentication keys and confirmed as active within the SIEM dashboard.

Real-time telemetry from both agents was validated through security events, MITRE ATT&CK mappings, and configuration assessments. The Windows Server agent also completed a CIS Benchmark scan with 130 passed checks, 209 failed checks, and a 38% compliance score, demonstrating the SIEM's ability to identify misconfigurations and enforce standards. Overall, the integration highlights the value of HIDS-to-SIEM communication for enterprise-level monitoring, threat detection, and compliance visibility across diverse environments.

Introduction

The purpose of this project is to demonstrate how a Host Based Intrusion Detection System (HIDS) agent is registered with a Security Information and Event Management (SIEM) console. HIDS agents collect endpoint telemetry such as process activity, file integrity changes, authentication events, and configuration data, which, when integrated with a SIEM, provide centralized visibility and real time threat detection. This project used Wazuh, an open source SIEM and HIDS platform, with a Wazuh Manager running on Ubuntu Server and a monitored Windows Server 2022 endpoint.

Environment Setup

Three virtual machines were used to complete this project:

Ubuntu Server

- Hosted the Wazuh Manager backend
- Provided centralized log collection and analysis
- Registered Agent 001 (Ubuntu Desktop) and 003 (Windows Server)

Ubuntu Desktop

- Hosted the Wazuh Dashboard GUI for SIEM access
- Also served as a monitored endpoint with Agent 001 installed
- Allowed browser-based access to the SIEM interface

Windows Server 2022

- Hosted the Wazuh HIDS agent (Agent 003)
- Served as the primary monitored endpoint for compliance and telemetry validation

All three systems were configured on the same virtual network to ensure direct communication between agents and the SIEM Manager.

Installing the Wazuh Manager (Ubuntu Server)

The Wazuh Manager and Dashboard were installed using the official Wazuh installation script. This included:

- Installing the Wazuh Manager service
- Installing the Wazuh API
- Installing the Wazuh Dashboard
- Verifying that all services were active

Once installed, the dashboard was accessible via HTTPS and ready to accept agent registrations.

```
11/12/2025 15:59:27 INFO: Generating configuration files.
11/12/2025 15:59:30 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passwords necessary for installation.
11/12/2025 15:59:30 INFO: --- Wazuh indexer ---
11/12/2025 15:59:30 INFO: Starting Wazuh indexer installation.
11/12/2025 16:01:39 INFO: Wazuh indexer installation finished.
11/12/2025 16:01:39 INFO: Wazuh indexer post-install configuration finished.
11/12/2025 16:01:39 INFO: Starting service wazuh-indexer.
11/12/2025 16:02:06 INFO: wazuh-indexer service started.
11/12/2025 16:02:06 INFO: Initializing Wazuh indexer cluster security settings.
11/12/2025 16:02:17 INFO: Wazuh indexer cluster initialized.
11/12/2025 16:02:17 INFO: --- Wazuh server ---
11/12/2025 16:02:17 INFO: Starting the Wazuh manager installation.
11/12/2025 16:06:28 INFO: Wazuh manager installation finished.
11/12/2025 16:06:28 INFO: Starting service wazuh-manager.
11/12/2025 16:06:52 INFO: wazuh-manager service started.
11/12/2025 16:06:52 INFO: Starting Filebeat installation.
11/12/2025 16:07:08 INFO: Filebeat installation finished.
11/12/2025 16:07:12 INFO: Filebeat post-install configuration finished.
11/12/2025 16:07:12 INFO: Starting service filebeat.
11/12/2025 16:07:14 INFO: filebeat service started.
11/12/2025 16:07:14 INFO: --- Wazuh dashboard ---
11/12/2025 16:07:14 INFO: Starting Wazuh dashboard installation.
11/12/2025 16:12:03 INFO: Wazuh dashboard installation finished.
11/12/2025 16:12:03 INFO: Wazuh dashboard post-install configuration finished.
11/12/2025 16:12:03 INFO: Starting service wazuh-dashboard.
11/12/2025 16:12:05 INFO: wazuh-dashboard service started.
11/12/2025 16:12:41 INFO: Initializing Wazuh dashboard web application.
11/12/2025 16:12:41 INFO: Wazuh dashboard web application not yet initialized. Waiting...
11/12/2025 16:12:58 INFO: Wazuh dashboard web application not yet initialized. Waiting...
11/12/2025 16:13:13 INFO: Wazuh dashboard web application initialized.
11/12/2025 16:13:13 INFO: --- Summary ---
11/12/2025 16:13:13 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
User: admin
Password: YAxMn.c.g6eYFRUR5c8+10WxNWrHj9as
11/12/2025 16:13:13 INFO: Installation finished.
turjak143@ubuntuServer:~$ S_
```

Figure 1: Wazuh Manager installation log confirming successful setup of all components.

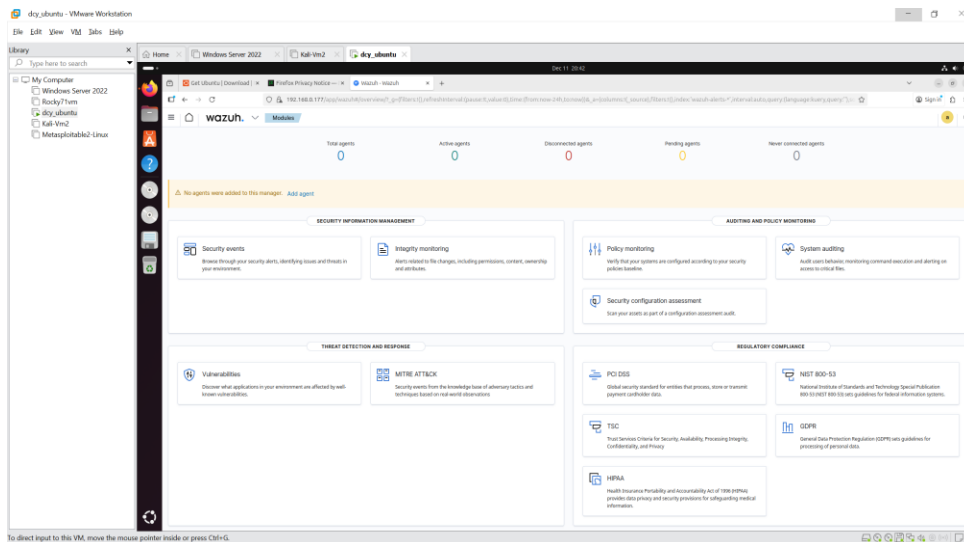


Figure 2: Wazuh Dashboard successfully loaded in Ubuntu Desktop Firefox browser, confirming SIEM interface is operational.

Installing the Windows HIDS Agent

The Wazuh Agent MSI installer was downloaded and executed on the Windows Server. During installation:

- The Manager IP was set to 192.168.0.177
- The agent service was installed
- The Wazuh Agent Manager application was made available

After resolving a corrupted service entry and reinstalling the agent, the service started successfully

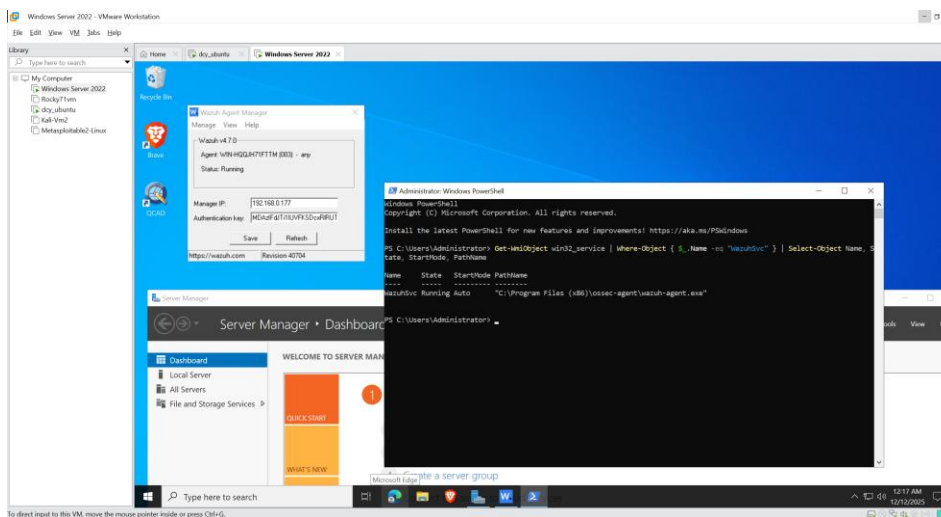


Figure 3: Wazuh Agent Manager and PowerShell output confirming agent 003 installation, Manager IP, and running service.

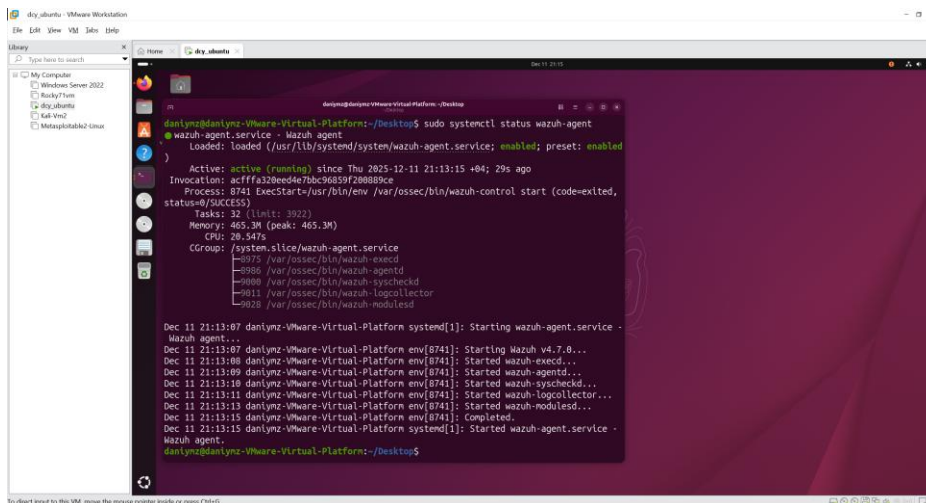


Figure 4: Ubuntu Desktop terminal confirming successful installation and active status of Wazuh Agent 001.

Registering the Agent with the SIEM

Agent registration required generating an authentication key on the Wazuh Manager using:

`sudo /var/ossec/bin/manage_agents`

Steps included:

- Adding the agent on the Ubuntu Manager
- Extracting the agent key
- Importing the key into the Windows Agent Manager
- Restarting the agent service
- Accepting the agent in the Wazuh Dashboard

Once accepted, the agent appeared as active.

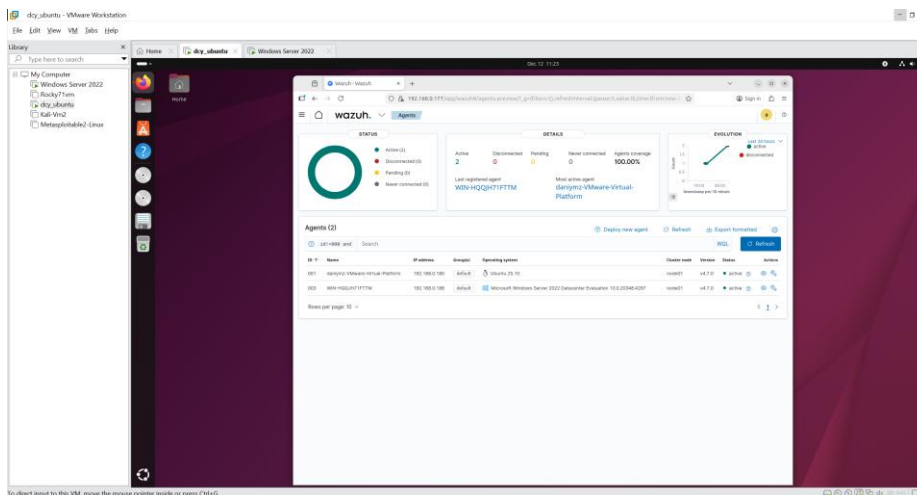


Figure 5: Wazuh Dashboard showing accepted Windows agent with active status and telemetry graphs.

Validating Agent Communication

To confirm successful communication, the Wazuh Dashboard was used to view:

- Real time logs
- Security events
- MITRE ATT&CK mappings
- Compliance scan results

The agent generated telemetry visible under:

Security Events → *Events*

This confirmed that the agent was actively sending logs to the SIEM.

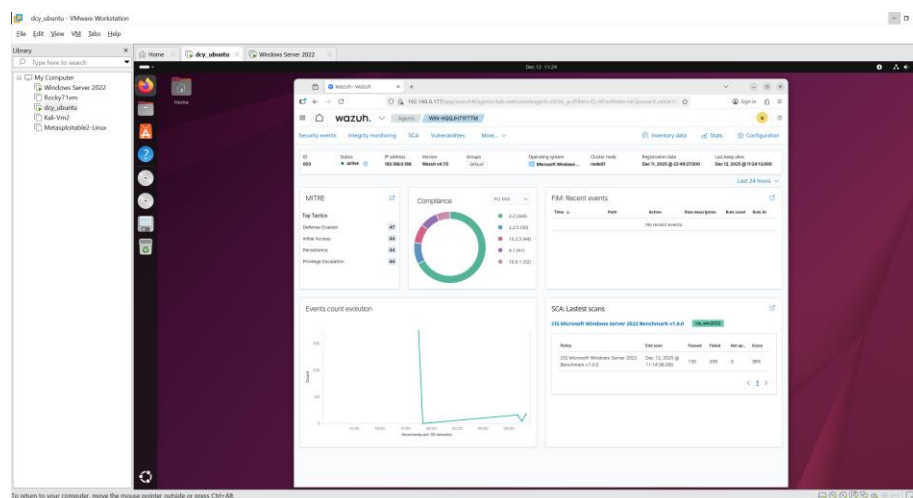


Figure 6: Wazuh Dashboard showing recent security events, MITRE tactics, and compliance telemetry from the registered agent.

Security & Compliance Insights

The SIEM automatically performed several security assessments:

MITRE ATT&CK

The dashboard mapped detected behaviors to MITRE tactics such as:

- Defense Evasion
- Initial Access
- Persistence
- Privilege Escalation

Compliance (CIS Benchmark)

The agent completed a CIS Microsoft Windows Server 2022 Benchmark scan, producing:

- 130 Passed checks
- 209 Failed checks
- Overall score: 38%

This demonstrates the SIEM's ability to identify misconfigurations and compliance gaps.

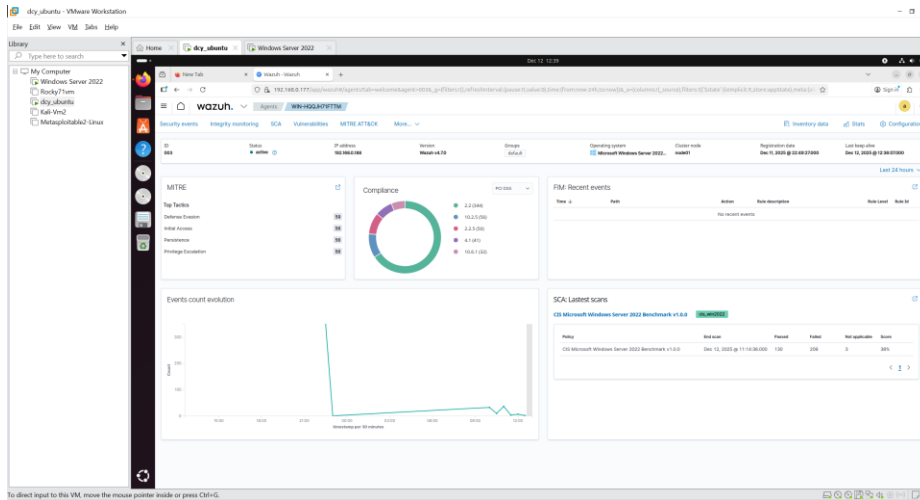


Figure 7: Wazuh Dashboard displaying MITRE tactics and CIS Benchmark scan results for Windows Server agent.

Conclusion

This project successfully demonstrated the deployment and registration of HIDS agents with a Wazuh-based SIEM across a three-machine environment. The Ubuntu Server hosted the Wazuh Manager, the Ubuntu Desktop provided the Dashboard GUI and ran Agent 001, and the Windows Server hosted Agent 003.

Both agents were installed, registered, and verified through real-time telemetry, MITRE ATT&CK mappings, and security configuration assessments. The Windows Server agent also completed a CIS Benchmark scan, highlighting configuration strengths and weaknesses.

With both Linux and Windows endpoints reporting to the SIEM, the environment now delivers centralized visibility, improved threat detection, and actionable compliance insights suitable for enterprise-level security operations.

References

Wazuh Documentation: [Wazuh documentation](#)

IT312.001 - Cybersecurity Foundations Course:

<https://byupw.instructure.com/courses/29353/files/4215926?wrap=1>

Ubuntu Installation guide: [Install Ubuntu Desktop | Ubuntu](#)

Windows Server installation guide: [Install Windows Server from installation media | Microsoft Learn](#)